

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
4	(a)			$\text{Mg} + \text{HCl} \longrightarrow \text{MgCl}_2 + \text{H}$ ☐ $\text{Mg} + 2\text{HCl} \longrightarrow \text{MgCl}_2 + 2\text{H}$ ☐ $\text{Mg} + 2\text{HCl} \longrightarrow \text{MgCl}_2 + \text{H}_2$ ☒		1		1		
	(b)	(i)		award (2) for all points plotted correctly – tolerance ± 1 square award (1) for any three points plotted correctly award (1) for (smooth) curve drawn through points judgement by eye ecf possible from incorrectly plotted points		3		3	3	
		(ii)		decreases increases			2	2		
		(iii)		more (1) a greater (1)	2			2		

Question				Marking details		Marks available					
						AO1	AO2	AO3	Total	Maths	Prac
		(iv)		Increasing the temperature of the acid	☒						
				Using a lump of magnesium	☐						
				Using a different apparatus	☐			2	2		2
				Using magnesium powder	☒						
				Decreasing the temperature of the acid	☐						
				Question 4 total		2	4	4	10	3	2